

Guan Chen

Summary

Platform engineer with 13+ years building distributed data infrastructure, transitioning from scientific computing to enterprise platform engineering. Designed PB-scale analytical platforms, streaming systems at 10B+ events/day and 1M+ connected vehicles, and privacy computing execution pipelines with query planning and storage abstraction.

Experience

Senior Data Platform Engineer

Shanghai Puxin Future Internet Research Institute | Dec 2024 – Present

MIRA / ChainWeaver — Privacy Computing Platform

- Built Data Service and storage abstraction supporting 10+ heterogeneous data sources, billion-scale table reads, and S3 object storage access
- Designed PQL execution planning pipeline converting declarative queries into distributed DAGs, enabling application teams to submit queries instead of custom orchestration scripts
- Architected query planner separating query semantics from execution strategy, allowing execution engine evolution without upstream contract changes
- Built reusable Data Service and storage abstraction layers, enabling new storage backends to integrate through stable interfaces
- Established platform interfaces and engineering standards across planner, scheduler, execution engine, and data service components
- Built benchmark platform to drive execution planning and runtime optimization with measurable evidence

Senior Data Platform Engineer / Technical Architect

Shanghai Jiayu Intelligent Robotics | Jun 2020 – Sep 2024

Connected Vehicle Telematics — National-Scale Streaming Platform (China)

- Scaled Apache Flink streaming platform from 10,000 to 1,000,000+ connected vehicles with stable production output on a 1000+ vCore cluster
- Owned end-to-end platform architecture processing 10B+ telemetry events/day
- Designed Kafka ingestion boundary decoupling millions of vehicle T-Box producers from downstream stream processing and storage consumers
- Built Flink pipelines for real-time aggregation and enrichment; Apache Doris serves as real-time warehouse for business-facing analytical queries
- Established metadata and data governance as platform-wide standards across national-scale telemetry datasets
- Integrated Dinky for Flink job lifecycle management across the 1000+ vCore cluster

Senior Data Platform Engineer

Ping An Securities | May 2017 – Apr 2020

Enterprise Financial Analytics — Metric Platform (China)

- Reduced 1000+ vCore batch computation time from 2.5 hours to 1.5 hours through Hive query optimization and Hadoop workload tuning
- Architected metric platform unifying PB-scale Hive analytics with Elasticsearch serving 100,000+ searchable metric fields to business applications without direct Hive access
- Designed dynamic schema expansion mapping 2000+ Hive physical columns to Elasticsearch through a Map-based translation layer
- Optimized Hive and Hadoop on 1000+ vCore cluster across PB-scale storage
- Established compute-serve separation pattern: batch computation in Hive, interactive search in Elasticsearch

Data Platform Engineer

Shanghai Jiayin Data Technology | Restaurant Intelligence / Wi-Fi Analytics

- Built streaming and batch pipelines for restaurant intelligence and commercial Wi-Fi analytics using Storm, Hive, and Hadoop

Research Engineer

Shanghai Ship and Shipping Research Institute | Scientific Computing

- Applied Hadoop and Hive to large-scale hydrodynamic simulation workloads

Skills

Platform: Distributed Systems, Execution Planning, Storage Abstraction, Metadata, Data Governance

Streaming & Compute: Kafka, Flink, Spark, Hive, Hadoop

Storage & Search: HDFS, Doris, Hudi, Elasticsearch, S3

Infrastructure: Linux, Docker, Kubernetes

Languages: Java, Python, C

Education

Master, Naval Architecture and Ocean Engineering — Shanghai Ship and Shipping Research Institute

Bachelor, Hydraulic Engineering — Tianjin University

Portfolio

Architecture documents, system design deep-dives, and interview preparation materials:

<https://github.com/jlty258/portfolio>